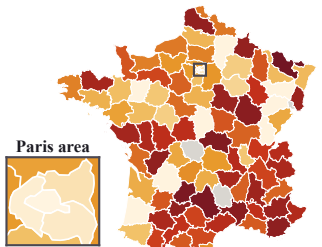
Do SHAP attributions agree with causal estimates?

Injury severity of e-scooter, e-bike, and bicycle riders in police-reported crashes, France, 2020–2025

46,920 riders | **44,817 crashes** | **96 départements** | **13 pre-specified factors**

DATA: RECORDED SEVERITY VARIES BY PLACE



Bicycle
30,040 riders
27.6% severe, unadjusted

E-bike
3,926 riders
26.8% severe, unadjusted

E-scooter
12,954 riders
18.6% severe, unadjusted

5% — 60%
killed or hospitalized
Paris 4.9% · highest 67.7%

METHOD

Prediction
gradient boosting

SHAP
attributions

developed on 2020–2023, tested once on 2024–2025

Causal estimation
double machine learning

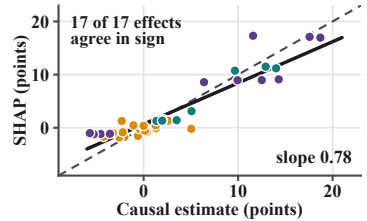
adjustment sets fixed by the temporal order of variables

Audit
direction · magnitude · rank · sources

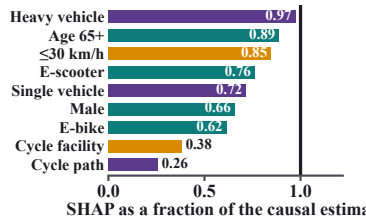
12 robustness checks, placebo, sensitivity analysis

FINDINGS

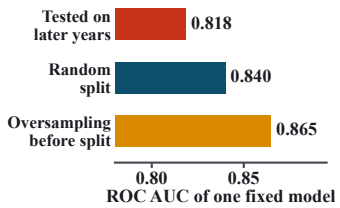
1 Same direction, smaller size



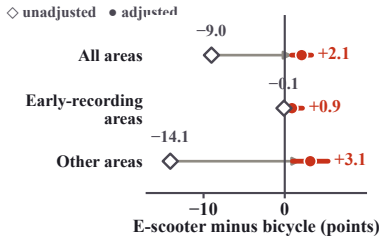
2 SHAP understates the effect



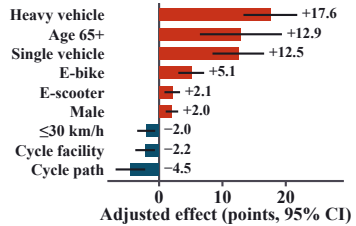
3 Shortcuts inflate accuracy



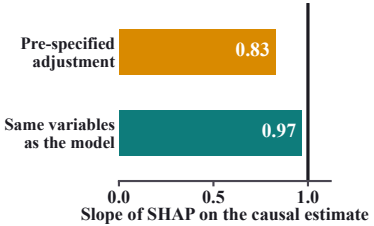
4 E-scooter advantage vanishes



5 What raises or lowers severity



6 Aligned adjustment closes the gap



For analysts

SHAP gives the direction of an effect, not its magnitude

For model builders

Test on later years and check whether recording depends on the outcome

For safety statistics

Compare modes only after adjusting for place; severity labels vary by area

For policy

Separate riders from heavy vehicles and address single-vehicle crashes